



Q. I am designing some LED lighting applications for my company. I do not know where to start. Can you please give me some pointers?

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A. LED lighting design looks very simple at first sight but you need to look at a lot of complex electrical, electronics and optics concepts to design one successfully. Each LED lighting system comprises five major subsystems: LEDs, electronics, optics, cooling system and mechanics.

LEDs. You first need to start with selecting the right LEDs for your application. The following are the specifications that you need to study carefully for each LED and decide the most suitable LED for you.

Correlated colour temperature (CCT). This is an important specification and, if you are designing the general lighting

application, this is the first thing you should find in the system requirement specifications document from the customer. White light is actually a mixture of colours and not all white LEDs are with the same proportionate combination of colours. So a white light with a higher proportion of red will appear warmer and a white light with a higher proportion of blue will appear cooler. In order to classify the different types of white lights, the concept of colour temperature is applied, which is described as the colour impression of a perfect black-body radiator at certain temperatures.

Colour rendering index (CRI). Two lights that appear to be the same may be made of different blends of wavelengths. As a result, different objects may appear differently since their surface may not reflect the same constituent wavelength. The value for CRI can also be seen as a comparison with the natural light. Refer Table I for CRI values.

Lumens. Check out overall lumens requirement for the project. This will help you decide on the LEDs and the number of LEDs you should be using to achieve the required lumens. Also check the system requirement specifications document for overall wattage of the product. With this wattage, you can decide on the efficacy of the LED you need.

Electronics. Once the LEDs are decided, the next step will be to design the electronics to drive them. You would need an AC/DC or DC/DC power supply to provide the required voltages and LED drivers to provide constant current to the LEDs. Check

datasheets of the LEDs for their driving current, which will help you decide on the LED drivers and the power supply.

Optics. The most critical part of the LED lighting is optics design. Decide on the broad architecture of LED light. Depending on the overall structure, you would need three optical components: light guide, reflector and diffuser. The specifications of these components will depend on the project at hand. Study these three components and finalise the specifications based on your requirement.

Cooling system. The performance of an LED is dependent on its temperature during operation. The design of the luminaire will influence its operating temperature and hence published characteristics. The paint finish/colour may affect the heat dissipation from the luminaire. A PCB is the interface between an LED and the heat sink and has a thermal resistance value. The higher the resistance the less efficient the system is at conducting away heat from the LED. This may impact the LED lumen output performance and ultimately its life.

Mechanics. Do check the system requirement specifications document for IP rating. The mechanical integrity of a luminaire is important in different areas including IP rating. The best thing for an LED lighting application is to have an integrated heat sinking capability of mechanics. Many LEDs can react to different chemicals, especially to phosphor. Some gaskets can out-gas chemicals which can affect the performance of some LEDs.

Standards. The IEC has published two publicly-available specification (PAS) performance requirement documents dealing with the LED products:

1. IEC/PAS 62717 performance requirements—LED modules for general lighting

2. IEC/PAS 62722 performance requirements—LED luminaires for general lighting

Some Indian standards are also available. Check Table II for Indian Standards.

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/A, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyjindia.com) and should include name and address of the sender

TABLE I
CRI Values

Code	CRI range	Colour rendering properties
6	57-66	Poor
7	67-76	Moderate
8	77-86	Good
9	87-100	Excellent

TABLE II
Some Indian Standards

Code	Standard
16101: 2012	General lighting—LEDs and LED modules—terms and definitions
16102 (Part 1): 2012	Self-ballasted LED lamps for general lighting services
16102 (Part 2): 2012	Self-ballasted LED lamps for general lighting services
16103 (Part 1): 2012	LED modules for general lighting
16103 (Part 2): 2012	LED modules for general lighting
15885 (Part2/Sec13): 2012	Safety of lamp control gear
16105: 2012	Method of measurement of lumen maintenance of solid-state light (LED) sources
16106: 2012	Method of electrical and photometric measurements of solid-state lighting (LED) products
10 16107 (Part 1): 2012	Luminaires performance part 1 general requirements
11 16107-1: 2012	Luminaires performance part 2 particular requirements